



Original Research



The Potential of Parasites in Grouper Fish (Family Epinepheliidae Bleeker, 1874) as Bioindicators of Water Quality at Kedonganan Beach, Bali

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ABSTRACT

The presence of parasites in the host is sensitive to changes in the aquatic environment, making them useful as bioindicators of water quality. Their reliance on the life cycle and environment of their hosts makes parasites sensitive to the health of aquatic ecosystems. The study examined 40 Epinepheliidae grouper samples from five species: *Cephalopholis miniata*, *Epinephelus longispinis*, *Epinephelus merra*, *Epinephelus ongus*, and *Epinephelus tauvina*. The fish were obtained from fishermen who caught and landed them at Kedonganan beach. Parasites infecting the five grouper species included *Rocinela* sp., *Acanthocephalus* sp., and *Lecithochirium* sp. The ratio of ectoparasites to endoparasites was 1:2, indicating that endoparasites are more common and suggesting that Kedonganan waters remain relatively stable. The prevalence ranged from 10% to 77.5%, falling into the normal-to-very-frequent category, while the intensity of infection was between 1 and 1.25 parasites per fish, categorized as low. The diversity index (H') ranged from 0.16 to 0.36, reflecting low diversity, and the Dominance Index (D) ranged from 3.0×10^{-4} to 1.5×10^{-1} , indicating low to very low dominance. The low diversity and dominance indices are likely due to a decrease in the number of intermediate hosts.

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INTRODUCTION

Kedonganan Beach is one of the beaches located in the Jimbaran area of Bali and is famous for its culinary tourism featuring fish caught by fishermen in the surrounding waters. The fish caught in the waters of Kedonganan are dominated by fish of significant economic value, including grouper (*Epinephelus* sp.), snapper (*Lutjanus* sp.), lemuru (*Sardinella lemuru*), and bullet tuna (*Euthynnus affinis*) (Ministry of Maritime Affairs and Fisheries, 2024). Grouper has good prospects in both domestic and international markets (FAO, 2025). However, fishermen's catches often decline in quality due to

fish diseases and parasites (Nastiti *et al.*, 2024), thereby affecting the selling price of groupers. Fish parasites are divided into ectoparasites and endoparasites (Sindermann, 1990s)

Ectoparasites include ciliates, flagellates, monogeneans, copepods, isopods, branchiurans, and leeches, whereas endoparasites infect the host's internal organs (Poulin *et al.*, 2008). The presence of parasites in hosts (fish) is sensitive to environmental conditions (Kloneirtz *et al.*, 2022); therefore, their presence can be used as a bioindicator of water quality (Palm and Ruckert, 2022). The direct relationship between parasites' life cycles and their



host environments makes them sensitive indicators of aquatic ecosystem health (Overstreet, 1997). According to Sures (2001) and Sures (2003), Acanthocephala parasites can be used to detect heavy metal pollution because they accumulate 1000 times more heavy metals than their host tissue. Vidal-Martinez *et al.* (2010) stated that digenean parasites can be used as aquatic indicators because heavy metals, such as lead (Pb), can inhibit their life cycles. Parasites with a direct life cycle, such as Trichodinid ciliates, can thrive in polluted water and indicate high bacterial counts (Ogut and Palm, 2005). Furthermore, the prevalence of Trichodinid ciliates indicates that the water is enriched with bacteria (Palm and Ruckert, 2009). The abundance and distribution of parasites on fish are influenced by body size and the fishing season (Mendes *et al.*, 2026).

Endoparasites have complex life cycles and tend to live in stable, unpolluted waters (Lafferty *et al.*, 2008). Changes in their abundance and diversity may reflect broader changes in aquatic ecosystem structure. Recent field evidence shows that water quality can alter both ectoparasite and endoparasite diversity and infection patterns in fish (Latief *et al.*, 2024). Therefore, evaluating parasites of Epinepheliidae as potential bioindicators at Kedonganan Beach may provide complementary information for monitoring ecosystem condition and supporting sustainable coastal management. Furthermore, the parasite count/ratio in groupers also indicates that the aquatic environment is still relatively stable or unstable.

MATERIALS AND METHODS

Time and Place of Research

This research was conducted over a period of 6 months to collect samples of grouper fish from fishermen who landed their catch on Kedonganan Beach in the morning. A total of 40 fish samples belonging to the Epinepheliidae family were collected for this study.

Tools and Materials

The tools used in this study included a stereo microscope, a binocular microscope, a dissecting set, a tray, an object glass, a cover glass, a Petri dish, and a spray bottle. The materials used included alcohol (70%, 85%, and 96%), distilled water, 0.9% NaCl, Canada balsam, and acetocarmine.

Research Design

This study used a quantitative descriptive method to obtain a systematic, factual, and accurate description or overview (Anwar, 2011). A quantitative descriptive method was used to describe the condition of the Kedonganan waters based on parasite counts.

The sampling method used was purposive sampling, a technique for selecting samples based on specific criteria (Etikan and Bala, 2017). This method was used to determine the sampling of grouper fish, which were obtained directly from

fishermen who caught the fish and landed them on the coast of Kedonganan.

Research Data Analysis

Prevalence

Prevalence is the percentage of fish infected with parasites in the fish population. The formula for calculating prevalence (Kabata, 1985) is as follows:

$$\text{Prevalence} = \frac{N}{n} \times 100 \%$$

Information:

N= Number of infected fish

n = Number of fish examined

Intensity

Intensity describes the density of parasites that can infect fish and can be calculated using the following formula (Kabata, 1985):

$$\text{Intensity} = \frac{P}{n}$$

Information:

P = Number of parasites infecting fish

n = Number of fish infected with parasites

Diversity Index (H')

The parasite diversity index is a formula used to calculate the diversity of parasite species found in fish (Odum, 1971).

$$(H') = \sum P_i \ln P_i$$

Information:

H'= Diversity

Pi = Number of parasite individuals found

Dominance Index (D)

The Dominance Index (D) is an ecological parameter that measures the level of parasite dominance in an area (Odum, 1971).

$$D = \sum \left(\frac{n_i}{N} \right)$$

Information:

Ni= Number of parasite individuals found

N= Total species of parasite found

RESULTS AND DISCUSSIONS

Grouper Fish Family Epinepheliidae Bleeker, 1874

The fish samples collected for parasite analysis belonged to the Epinepheliidae family, consisting of 5 species, namely *Cephalopholis miniata* (7 specimens), *Epinephelus longispinis* (6 specimens), *Epinephelus ongus* (12 specimens), *Epinephelus tauvina* (10 specimens), and *Epinephelus merra* (5 specimens). Collect a sample from Kedonganan Beach.

Parasites Infecting Grouper Fish (Epinepheliidae Bleeker, 1874)

This study found three parasite species infecting Epinepheliidae: *Rocinela* sp., *Acanthocephalus* sp., and *Lecithochirium* sp., with a total of 71 individuals (Table 1).

Table 1. Types and numbers of parasites infecting Epinepheliidae. Based on the research results, three parasites

No.	Species Grouper	Ectoparasite	Endoparasite		Total
		<i>Rocinela</i> sp.	<i>Acanthocephalus</i> sp.	<i>Lecithochirium</i> sp.	
1.	<i>Cephalopholis miniata</i>	3	-	9	12
2.	<i>Epinephelus longispinis</i>	4	-	6	10
3.	<i>Epinephelus ongus</i>	13	3	14	31
4.	<i>Epinephelus tauvina</i>	7	-	10	17
5.	<i>Epinephelus merra</i>	1	1	-	2
Parasite Total		28	4	39	71



Figure 1. Parasites were found in Epinepheliidae. Description: a) *Rocinela* sp.; b) *Acanthocephalus* sp.; c) *Lecithochirium* sp.

Based on the parasites infecting them, *Epinephelus ongus* was the fish species infected by all three parasites, with the highest number of infections among them species were found infecting five grouper species: *Cephalopholis miniata*, *Epinephelus longispinis*, *Epinephelus merra*, *Epinephelus ongus*, and *Epinephelus tauvina*: *Rocinela* sp., *Acanthocephalus* sp., and *Lecithochirium* sp., totaling 71 individuals, comprising 28 ectoparasites and 43 endoparasites. The +/- ectoparasite/endoparasite ratio method was used to determine the water quality in Kedonganan. The ratio of ectoparasites to endoparasites was 1:2, based on the number of species identified. The parasite was identified using a binocular microscope (Figure 1).

The ectoparasite-endoparasite ratio was interpreted as a complementary ecological indicator because these parasite groups differ in transmission pathways and sensitivity to host and environmental change (Palm and Bray, 2014; Kleinertz et al., 2022). In this study, endoparasite taxa outnumbered ectoparasite taxa at a ratio of 2:1. This pattern is consistent with relatively intact trophic transmission pathways, but it should be interpreted together with other parasitological indices rather than as a stand-alone measure of water quality. Endoparasites inhabit internal organs and frequently have complex life cycles involving one or more intermediate hosts, including mollusks, crustaceans, or other prey organisms. Their successful transmission therefore depends on the availability of suitable hosts and on trophic interactions within the ecosystem (Kleinertz et al., 2022).

Accordingly, reductions in intermediate-host abundance or disruption of food-web links can reduce transmission of trophically transmitted endoparasites. Environmental contaminants may also modify parasite development and host-parasite

interactions, so parasite patterns should be interpreted in relation to local ecological context (Sures et al., 2017; Sures and Nachev, 2022). The occurrence of endoparasites may therefore provide information on the persistence of intermediate hosts and trophic links. In the present study, *Lecithochirium* sp. was common in grouper, a pattern consistent with continuing transmission through prey organisms that act as intermediate or paratenic hosts (Kleinertz et al., 2022).

Lecithochirium sp. was found in 31 of the 40 fish examined, with 39 individuals recovered. This comparatively high occurrence suggests that transmission opportunities for this digenean remain available in the Kedonganan food web; however, confirmation of the specific intermediate hosts in this locality would require direct sampling of mollusks and crustaceans. Ectoparasites live on the external parts of fish, such as the skin, fins, and gills, and are generally opportunistic feeders. Increased ectoparasite populations are often associated with poor water quality (Bustos et al., 2020). Increased pollutants (such as industrial waste, heavy metals, and agricultural chemicals), decreased dissolved oxygen, and extreme temperature fluctuations can cause stress in fish.

Environmental stress can alter host immunity and parasite transmission, but the direction of the response varies among parasite groups and pollutants. Therefore, increases or decreases in ectoparasite abundance should not be interpreted as a universal pollution signal without supporting water-quality data (Sures et al., 2017; Latief et al., 2024).

In this study, 26 fish were infected with the ectoparasite *Rocinela* sp., with a total of 28 parasites. According to Bharadhirajan et al. (2014), *Rocinela* sp. parasites feed on blood and are commonly found on the gills of marine fishes. This activity injures the

fish, making the host susceptible to secondary infections and reducing its growth rate. *Rocinela* sp. species are ectoparasites in the class Isopoda, which generally infect by damaging gill tissue. According to Bahtiar *et al.* (2024), isopods obtain their food from the blood capillaries in their gills.

Acanthocephalans are commonly known as thorny- or spiny-headed worms because the anterior proboscis bears hooks used for attachment to the intestinal wall. They lack an alimentary canal and absorb nutrients across the body surface. Their life cycles typically include an arthropod intermediate host, in which the infective cystacanth develops before transmission to the definitive vertebrate host (Minnot *et al.*, 2023). In this study, adult *Acanthocephalus* sp. were recovered from the grouper intestine.

Prevalence

The prevalence in this study ranged from 10% to 77.5%, with categories from common to very frequent. The species with the lowest prevalence in this study was *Acanthocephalus* sp. (10%) in the common category, *Rocinela* sp. (65%) in the frequent category, and the highest prevalence was for *Lecithochirium* sp. (77.5%) in the very frequent category. *Epinephelus ongus* was infected with the three parasites with the highest case counts, at 31 individuals. Thirteen *Rocinela* sp. infections were found in the gills and operculum, three *Acanthocephalus* sp. infections infected the intestines, and 14 *Lecithochirium* sp. infections were found in the intestines and stomach of the fish.

Parasite prevalence varied among taxa, reflecting differences in transmission pathways, host exposure, and host-parasite specificity. *Lecithochirium* sp. showed the highest prevalence, whereas *Acanthocephalus* sp. showed the lowest. Such variation is expected because host diet, size, habitat use, and the availability of intermediate hosts can influence parasite community structure and infection levels (Koepper *et al.*, 2020; Pravdova *et al.*, 2023). The prevalence of parasites in this study ranged from 10% to 77.5%, with prevalence ranging from common to very frequent. *Acanthocephalus* sp. was 10%, with prevalence ranging from common to very frequent, from 65% to very frequent, and from 77.5% to very frequent. Several interacting factors cause the observed variation in parasite prevalence. These factors include the parasite's host specificity (Sures *et al.*, 2017). Some parasites, such as *Lecithochirium* sp., may have very common, readily found definitive hosts in the study area. Meanwhile, *Acanthocephalus* sp. may have more specific or rare hosts, resulting in a low prevalence (Radwan *et al.*, 2025). The high prevalence of the endoparasite *Lecithochirium* sp. relative to other parasites is due to the carnivorous diet of groupers, which feed on gastropods and sea snails (Serna *et al.*, 2019). According to Nachev and Sures (2022), carnivorous fish are more susceptible to infection by endoparasites, such as *Lecithochirium* sp. and *Anisakis* sp., because their diets overlap with the life cycles of digenetic and nematode endoparasites,

which use crustaceans and gastropods as intermediate hosts, making them more susceptible to infection.

This is supported by Kleinertz *et al.* (2022), who stated that fish that consume gastropods, sea snails, and crustaceans are more susceptible to infection by Digenea and nematodes, as these taxa are the primary intermediate hosts for these parasites. The Prevalence graph shown in Figure 2.

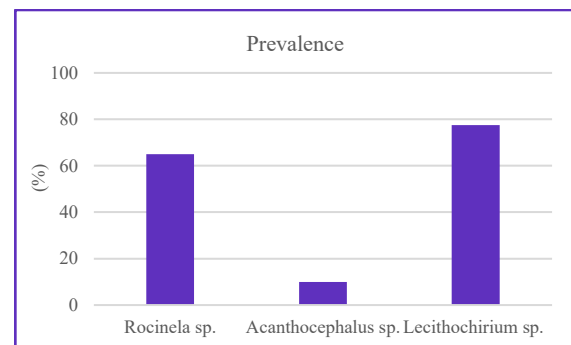


Figure 2. Prevalence of Parasite from Epinepheliidae

Intensity

The intensity value in this study ranged from 1 to 1.25 ind/fish, which is a low category. The lowest intensity was *Acanthocephalus* sp. at 1 ind/fish, followed by *Rocinela* sp. at 1.07 ind/fish and *Lecithochirium* sp. at 1.25 ind/fish. The intensity values ranged from 1 to 1.25 ind/fish and were categorized as low. The lowest intensity was observed for *Acanthocheilus* sp. at 1 ind/fish, followed by *Rocinela* sp. at 1.07 individuals/fish and *Lecithochirium* sp. at 1.25 ind/fish. These values indicate several factors, including the influence of grouper size, which ranged from 18 to 24.6 cm and was considered medium-sized. According to Poulin (2000), fish size can influence parasite intensity, as greater length and weight can influence parasite intensity. Medium-sized fish have strong immune systems and can effectively fight parasitic infections compared to smaller fish; therefore, the number of parasites that survive in their bodies is very low (Khammassi, 2024). Good, unpolluted water quality likely eliminates stressors that can weaken the host's immune system (Sures *et al.*, 2025). A healthy environment can break the chain of parasite transmission, thereby reducing host exposure (Grabner *et al.*, 2023). The intensity graph is shown in Figure 3.

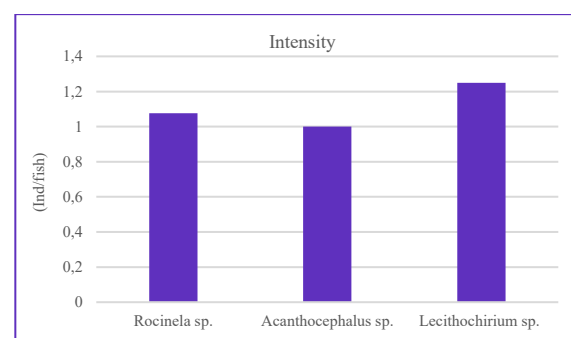


Figure 3. Intensity Parasite from Epinepheliidae

Diversity (H') and Dominance Index (D)

The Diversity Index (H') from this study ranged from 0.16 to 0.36, with the lowest Diversity Index value found in *Lecithochirium* sp. and the highest in *Rocinela* sp. The Dominance Index (D) of each parasite ranged from 3.0×10^{-4} to 1.5×10^{-1} , indicating a low dominance category: *Rocinela* sp. and *Acanthocephalus* sp. Diversity Index (H') values ranged from 0.16 to 0.36, categorized as low, while dominance (D) values ranged from 3.0×10^{-4} to 1.5×10^{-1} , categorized as low (*Rocinela* sp. and *Acanthocephalus* sp.) and very low (*Lecithochirium* sp.). This may be due to a reduction in the number of potential first intermediate hosts, which is a crucial stage in the parasite's life cycle (Palm *et al.*, 2020). The intermediate host of *Acanthocephalus* sp. is a crustacean (Minnot *et al.*, 2023), whereas those of *Lecithochirium* sp. are gastropods and crustaceans (Lafferty *et al.*, 2020). According to Koepper *et al.* (2020), seasons can influence the diversity and dominance of fish parasites by affecting the availability of intermediate hosts. This study was conducted during the dry season, which led to declines in several intermediate hosts, such as gastropods and crustaceans, thereby reducing parasite accumulation in fish. According to Tavares *et al.* (2015), the dry season can cause stress in intermediate hosts, disrupting the parasite life cycle and affecting its presence in fish. The table of diversity indices (H') is presented in Table 2, and the dominance index is shown in Table.

Table 2. Diversity Index (H')

Parasite Species	H'	Category
<i>Rocinela</i> sp.	0.36	Low
<i>Acanthocephalus</i> sp.	0.32	Low
<i>Lecithochirium</i> sp.	0.16	Low

Table 3. Dominance Index (D)

Parasite Species	Dominance Index (D)	Category
<i>Rocinela</i> sp.	1.5×10^{-1}	Low
<i>Acanthocephalus</i> sp.	3.0×10^{-1}	Low
<i>Lecithochirium</i> sp.	3.0×10^{-4}	Very Low

CONCLUSION

The parasite assemblage of grouper landed at Kedonganan Beach comprised one ectoparasite taxon and two endoparasite taxa, giving an ectoparasite-to-endoparasite taxon ratio of 1:2. Prevalence ranged from 10.0% to 77.5%, while infection intensity was low (1.00-1.25 individuals per infected fish). The Shannon diversity index (H') ranged from 0.16 to 0.36 and Simpson's dominance index (D) from 3.0×10^{-4} to 1.5×10^{-1} . Together, the predominance of endoparasites and the continued occurrence of trophically transmitted parasites are consistent with the persistence of intermediate hosts and trophic links in the Kedonganan ecosystem. However, because parasite-based indicators are influenced by host ecology, season, and pollutant type, these findings should be interpreted as complementary evidence of relatively stable

ecological conditions rather than as a stand-alone measure of water quality.

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